

ASSIGNMENT : 8

TITLE : Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement : Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives : To demonstrate the use of the super keyword for accessing parent class members.

Theory : The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Exercise :

1. Create an **Employee and Manager program** where Manager accesses Employee details using super keyword.

```
class Employee {
    String name;
    int id;

    Employee(String name, int id) {
        this.name = name;
        this.id = id;
    }
}

class Manager extends Employee {
    String department;

    Manager(String name, int id, String department) {
        super(name, id);
        this.department = department;
    }

    void display() {
        System.out.println("Name: " + super.name);
        System.out.println("ID: " + super.id);
        System.out.println("Department: " + department);
    }
}

public class Main {
    public static void main(String[] args) {

        Manager m = new Manager("Anushree", 101, "HR");
        m.display();
    }
}
```

Output

Generated files

```
Name: Anushree  
ID: 101  
Department: HR
```

2. Develop a **Vehicle Insurance System** where child insurance classes access parent vehicle information using the super keyword.

```
class Vehicle {  
    String vehicleNo;  
  
    Vehicle(String vehicleNo) {  
        this.vehicleNo = vehicleNo;  
    }  
}  
  
class Insurance extends Vehicle {  
    String policyNo;  
  
    Insurance(String vehicleNo, String policyNo) {  
        super(vehicleNo);  
        this.policyNo = policyNo;  
    }  
  
    void display() {  
        System.out.println("Vehicle No: " + super.vehicleNo);  
        System.out.println("Policy No: " + policyNo);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Insurance obj = new Insurance("MH12AB1234", "INS5678");  
        obj.display();  
    }  
}
```

Output

Generated files

```
Vehicle No: MH12AB1234
```

```
Policy No: INS5678
```

```
|
```

Outcome :

- Learned the use of the **super** keyword.
- Accessed parent class constructors and variables.
- Understood constructor chaining in inheritance.

Conclusion :

This assignment demonstrated the use of the **super** keyword in Java. It enables child classes to access parent class constructors, variables, and methods, ensuring proper inheritance, constructor chaining, and improved code organization.